Vikram Kamath Cannanure

Postdoctoral Fellow - Saarland University · HCI, ICTD, AI, Education

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I design *Thoughtful AI*: human-centered systems for reflection, care, and trust, built with communities across HCI and ICTD.

- **Research** — 18 full papers at top-tier HCI venues (CHI, CSCW, COMPASS, ICTD); systems, datasets, evaluation frameworks.
- **Impact** — AI systems deployed across the Global South; teacher chatbot in Côte d'Ivoire: 400+ teachers, 10,000+ students.
- **Funding** — $215,000+ in competitive grants, including a Gates Foundation Grand Challenges award.
- **Recognition** — 513 citations, h-index 12 (Google Scholar, 2026); Best Paper Award at ACM COMPASS.
- **Leadership** — Co-led *HCI Across Borders*, a global HCI community; mentored 25+ students; reviewed 80+ papers.

Research Experience

Saarland University — *Saarbrücken, Germany*

Postdoctoral Fellow — *Oct 2023 – present*

- Host: Prof. Ingmar Weber · Leading a research area at the intersection of HCI, ICTD, and generative AI
- Co-supervising PhD and master's students on socially grounded AI projects
- Co-PI on a $100,000 Gates Foundation grant to co-design a chatbot improving women's and youth health in rural India
- Instructor for the graduate seminar "AI and the Global South"
- Key publications: [P13, P12, P14, S4]

Education

Carnegie Mellon University — *Pittsburgh, USA*

PhD, Human-Computer Interaction — *2017–2023*

- Advisors: Prof. Tim Brown, Prof. Amy Ogan · Committee: Chinmay Kulkarni, Mayank Goel, Kentaro Toyama (U. Michigan)
- Dissertation: *DIA: Supporting Teacher Professional Development in Low Infrastructure Settings*
- Led multi-year fieldwork and chatbot deployments for teachers in rural Côte d'Ivoire
- Awarded $100,000 Fellowship from the Center for Machine Learning and Health (CMLH)
- Key publications: [P6, P7, P8, P9, P10, P15]

Carnegie Mellon University — *Pittsburgh, USA*

MS, Human-Computer Interaction — *2017–2021*

Carnegie Mellon University — *Pittsburgh, USA*

MS, Educational Technology and Applied Learning Science — *2014–2015*

National Institute of Technology Karnataka — *Karnataka, India*

B.Tech. — *2007–2011*

Grants and Awards ($215,000+ total)

2026	**Deans' Global Innovation Challenge, UT Knoxville (Co-PI; under review)**, *Engineering Sustainable Global Futures: Cross-University Learning in Rwanda* — **$50,000**	*Rwanda*
2026–	**Listen at Scale (Gram Vaani)**, *Voice-first AI for reproductive and mental wellness* — **$5,000**	*India*
2025–	**Gates Foundation Grand Challenges (Co-PI)**, *LLM wellness platform for rural India* — **$100,000**	*India*
2019	**PhD Fellowship**, Center for Machine Learning and Health, CMU — **$100,000**	*Pittsburgh, USA*
2023, 2025	**SIGCHI Development Grant**, HCI Across Borders (HCIxB) workshop at CHI — **$10,000**	
2019	**Best Paper Award**, ACM COMPASS	*Accra, Ghana*
2019	**Poster Spotlight**, ACM COMPASS	*Accra, Ghana*

Publications

- **Record** — 35+ peer-reviewed publications: 18 full papers at *CHI*, *COMPASS*, *ICTD*, *CUI*; 3 under review.
- **Metrics (2026)** — 513 citations, h-index 12, i10-index 17. [Google Scholar] [ORCID]
- **Author order** — first = primary intellectual lead; last = supervision and project leadership; * = equal contribution.

Journal and Conference Full Papers [P]

[P18] **Cannanure, V.K.**, Yinkfu, B., Bryan, D., Amin, M., Weber, I., *Teacher Professional Development on WhatsApp and LLMs: Early Lessons from Cameroon*. Accepted at Proceedings of the International Conference on Artificial Intelligence in Education (AIED 2026), Springer LNCS. [Link]

[P17] Behmanush, H., Akhtari, F., Weber, I., **Cannanure, V.K.**, *Designing Safe and Accountable GenAI as a Learning Companion with Women Banned from Formal Education*. Accepted at Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT 2026), ACM. [Link]

[P16] Wolf, S.*, Ogan, A.*, Jasińska, K.*, Kembou, S.*, **Cannanure, V.K.**, Whitehead, H., Corbett, J., Tanoh, F., Wortsman, B., Lim, J., Zhao, H., Kumar, A. *Impacts of Targeted Instruction with Mobile-based Support on Teachers and Students in Rural Côte d'Ivoire*. Accepted at American Educational Research Journal. [Link]

[P15] **Cannanure V.K.**, Ngoon T., Wolf S., Jasińska K., Brown T.X., Ogan A. *A Design-Based Research Approach to Bridge Teacher Aspirations and Goal-Setting*. *ACM Journal on Computing and Sustainable Societies*. [Link]

[P14] Zhu J., Kempermann M., **Cannanure V.K.**, Hartland A., Navarrete R., Carteny G., Braun D., Weber I. *Learn, Explore and Reflect by Chatting: Understanding the Value of an LLM-Based Voting Advice Application*. *ACM CUI 2025*, Waterloo, Canada. [Link]

[P13] Adhikari D.M., Hartland A., Weber I., **Cannanure V.K.**. *Exploring LLMs for Automated Generation and Adaptation of Questionnaires*. *ACM CUI 2025*, Waterloo, Canada. [Link]

[P12] Behmanush H., Akhtari F., Nooripoor R., Weber I., **Cannanure V.K.** *Online Learning and GenAI: Supporting Women's Aspirations Amid Socio-Political Instability in Afghanistan*. *ACM COMPASS 2025* [Link]

[P11] Verma R., Weber I., **Cannanure V.K.**. *Exploring AI in Education: Child-Computer Interaction in the Global South*. *ACM Journal on Computing and Sustainable Societies*. [Link]

[P10] **Cannanure V.K.**, Ngoon T., Wolf S., Jasińska K., Brown T., Ogan A. *Understanding the Longitudinal Impacts of a Chatbot to Facilitate a Virtual Community of Practice for Teachers in Rural Côte d'Ivoire*. *ACM COMPASS 2024*, Delhi, India. [Link]

[P9] Ngoon T., **Cannanure V.K.**, Wolf S., Jasińska K., Brown T., Ogan A. *"I believe I did not preach into the desert": Opportunities & Challenges in Scaling Teacher Mentorship through Mobile Technology in Rural Côte d'Ivoire*. *L@Scale 2024*, Atlanta, USA. [Link]

[P8] **Cannanure V.K.**, Ávila E., Ngoon T., Adji Y., Wolf S., Jasińska K., Brown T., Ogan A. *"We dream of climbing the ladder, for getting there, we have to do our job better": Designing for Teacher Aspirations in Rural Côte d'Ivoire*. *ACM COMPASS 2022*, Seattle, USA. [Link]

[P7] **Cannanure V.K.**, Brown T., Ogan A. *DIA: A Human-AI Hybrid Chatbot for Low-Resource Contexts*. *ICTD 2020*, Guayaquil, Ecuador. [Link]

[P6] **Cannanure V.K.**, Souvenir J., Adji Y., Apke H., Jasińska K., Brown T., Ogan A. *"I'm fine where I am, but I want to do more": Exploring Teacher Aspirations in Rural Côte d'Ivoire*. *ACM COMPASS 2020*, Guayaquil, Ecuador. [Link]

[P5] Madaio M., Yarzebinski E., **Cannanure V.K.**, Zinszer B., Seri A., Tanoh F., Hannon-Cropp J., Cassell J., Jasińska K., Ogan A. *Collective Support and Independent Learning with a Voice-Based Literacy Technology in Rural Communities*. *CHI 2020*, Hawaii, USA. [Link]

[P4] Madaio M., **Cannanure V.K.**, Yarzebinski E., Zasacky S., Tanoh F., Hannon-Cropp J., Cassell J., Jasińska K., Ogan A. *"You Give a Little of Yourself": Family Support for Children's Use of an IVR Literacy System*. *ACM COMPASS 2019*, Accra, Ghana. (Best Paper Award) [Link]

[P3] **Cannanure V.K.**, Brown T. *RASP-IVR: A Low-Cost Interactive Voice Response System*. *AfriCHI 2018*, Windhoek, Namibia. [Link]

[P2] Onafuwa D., Early K., Vyas N., **Cannanure V.K.**, Mankoff J. *Understanding the Information Needs of Prospective Tenants*. *ACM COMPASS 2018*, San Francisco, USA. [Link]

[P1] Bhat K.*, Thinayane M.*, Goldkind L., **Cannanure V.K.** *Critical Participatory Design: Reflections on Engagement and Empowerment in a Case of a Community-Based Organization*. *Participatory Design Conference (PDC 2018)*, Hasselt and Genk, Belgium. [Link]

Book Chapters [B]

[B1] Gautam A., Wong-Villacrés M., Gaytán Lugo L.S., **Cannanure V.K.**, Vannini S., Jimenez A., Tuli A., Reynolds-Cuéllar P., Pei L., Alvarado Garcia A., Badillo-Urquiola K., Prietch S.S., Densmore M., Banganayi T., Dray S. *Human-Computer Interaction Across Borders: Perspectives from the Global South*. In *The Human-Computer Interaction Handbook* (4th ed.), Julie A. Jacko (Ed.). CRC Press, 2026.

Short Papers and Posters [S]

[S5] **Cannanure V.K.**, Wolf S., Jasińska K., Brown T., Ogan A. *Enabling a Virtual Community of Practice Facilitated by a Chatbot for Rural Teachers*. *Proceedings of the 5th African Human-Computer Interaction Conference (AfriCHI 2025), Cairo, Egypt*. [Link]

[S4] Mensah E.E., **Cannanure V.K.**, Weber I. *Ethnic Diversity and Spatial Dynamics in Google Maps Reviews: Insights from a German Border City*. *ICWSM 2025* (poster). [Link]

[S3] **Cannanure V.K.**, Wolf S., Jasińska K., Brown T.X., Ogan A. *Applying Think-Aloud in ICTD: A Case Study of a Chatbot Use by Teachers in Rural Côte d'Ivoire*. *ICTD 2024*, Nairobi, Kenya. [arXiv]

[S2] Adhikari D.M., **Cannanure V.K.**, Hartland A., Weber I. *Exploring LLMs for Automated Pre-Testing of Cross-Cultural Surveys*. *ICTD 2024*, Nairobi, Kenya. [arXiv]

[S1] **Cannanure V.K.**, Chen X., Mankoff J. *Twist and Knock: A One-Handed Gesture for Smart Watches*. *Proceedings of the 42nd Graphics Interface Conference (GI 2016), Victoria, Canada*. [Link]

Workshop Papers and Lightly Reviewed Work [L]

[L4] **Cannanure V.K.**, Varghese D., Rivera-Loaiza C., Noor F., Das D., Jain P., Chang-Rizzo M., Wong-Villacrés M., Karusala N., Ahmed N., Till S., Akhigbe B., Dray S., Densmore M, Sturm C., Kumar N. (2023). *HCI Across Borders: Towards Global Solidarity*. CHI '23, Apr 2023, Hamburg, Germany.

[L3] **Cannanure V.K.**, Karusala N., Rivera-Loaiza C., Prabhakar A. S., Varanasi R. A., Tuli A., Gamage D., Noor F., Nemer, D., Das D., Dray S., Sturm C., Kumar, N. (2022). *HCI Across Borders: Navigating Shifting Borders at CHI*. CHI '22, Apr 2022, New Orleans, USA.

[L2] **Cannanure V.K.**, Gamage D., Sturm C., Winschiers-Theophilus H., Maestre, J. F., Karusala N., Reynolds-Cuéllar P., Kumar, N., *Workshop: Decolonizing HCI Across Borders*. CHI '21, Apr 2021, Yokohama, Japan.

[L1] Kumar N., **Cannanure V.K.**, Gamage D., Prabhakar A., Sturm C. *Workshop: HCI Across Borders and Sustainable Development Goals*. CHI '20, Apr 2020, Hawaii, USA.

Teaching Experience

AI and the Global South (Block Seminar) — *Saarland University*, *Summer 2025*
Lead Instructor and Course Creator
Student evaluation: 1.22 overall vs. department average 1.51 (lower is better)

Elements of Data Science and Artificial Intelligence — *Saarland University*, *Fall 2024*
Course Coordinator (Instructor: Prof. Ingmar Weber)

Programming Usable Interfaces (05-430/05-630) — *Carnegie Mellon University*, *Spring 2022*
Teaching Assistant, hybrid (Instructor: Prof. Scott Hudson)

User-Centered Research & Evaluation (05-410/05-610) — *Carnegie Mellon University*, *Spring 2021*
Teaching Assistant, remote (Instructor: Raelin Musuraca)

Mentoring

PhD Students (2)

*__Hamayoon Behmanush__: Saarland University, PhD: *Supporting Afghan women in learning programming* (P12, P17)
*__Ethel Mensah__: Saarland University, PhD: *Exploring structural inequality in Germany using POI data* (S4)

Thesis Committees (1)

Ahana Ghosh: MPI-SWS, PhD 2026: committee member

Master's Students (21)

*__Divya Mani Adhikari__: Saarland University, MS 2025: *Automated Survey Testing using LLMs* (P13, S2)
Prithika Acharya: CMU Pittsburgh, METALS 2022: *Teacher Training Project in Côte d'Ivoire*
Aastha Patel: CMU Pittsburgh, METALS 2022: *Teacher Training Project in Côte d'Ivoire*
Egide Kamuzinzi: CMU Africa, MSIT 2021: *Teacher Training Project in Côte d'Ivoire*
Baptiste Moreau-Pernet: CMU Pittsburgh, METALS 2021: *Teacher Training Project in Côte d'Ivoire*
Saumya Sharma: CMU Pittsburgh, MSIIPM 2020: *Simple Health Information System*
Lucie Uwimpaye: CMU Africa, MSIT 2021: *Simple Health Information System*
Eloísa Ávila: CMU Pittsburgh, MSPPM 2021: *Teacher Training Project in Côte d'Ivoire*
Sheila Mbadi: CMU Africa, MSIT 2021: *Simple Health Information System*
Patrick Manirakiza: CMU Africa, MSECE 2019: *M&E system for Rwanda Women's Network*
Solange Iyubu: CMU Africa, MSECE 2019: *M&E system for Rwanda Women's Network*
Sabin Sheja: CMU Africa, MSECE 2019: *M&E system for Rwanda Women's Network*

Justin Bigwi: CMU Africa, MSECE 2019: *Design of a Phone Call-Based Health Information System*
Gloria Umutesi: CMU Africa, MSECE 2020: *Design of a Phone Call-Based Health Information System*
Jocelyne Mukamisha: CMU Africa, MSIT 2020: *Design of a Phone Call-Based Health Information System*
Justin Souvenir Niweteto: CMU Africa, MSIT 2020: *Design of a Phone Call-Based Health Information System* (P6)
Fraterne Kagame: CMU Africa, MSIT 2020: *Design of a Phone Call-Based Health Information System*
Huguette Ingenere: CMU Africa, MSIT 2020: *French interview transcription for teachers in Côte d'Ivoire*
Marie Claudine Ingenere: CMU Africa, MSIT 2020: *French interview transcription for teachers in Côte d'Ivoire*
Nebiyou Yismaw: CMU Africa, MSECE 2018: *Digital and voice-based systems to support gender-based violence response*
Fabiola Uwashema: CMU Africa, MSIT 2018: *Digital and voice-based systems to support gender-based violence response*

Undergraduate Students (3)

*__Riti Verma__: Indian Institute of Technology, Delhi: *Exploring the role of AI in the Global South* (P11)
Lena Armstrong: University of Pennsylvania, Summer REU: *Teacher Training Project in Côte d'Ivoire*
Maya Greenholt: CMU Pittsburgh, Cognitive Science: *Teacher Training Project in Côte d'Ivoire*

*Mentored during my postdoctoral fellowship at Saarland University.

Professional Service

Chairing and Organizing

2026	**Papers Chair**, ACM COMPASS 2026	*Virtual*
2025	**Lead Mentor**, HCI Across Borders (HCIxB) workshop at CHI 2025	*Yokohama, Japan*
2024	**Lead Organizer**, HCI Across Borders (HCIxB) workshop at ACM COMPASS	*Delhi, India*
2023	**Lead Organizer**, HCI Across Borders (HCIxB) workshop at CHI	*Hamburg, Germany*
2022	**Lead Organizer**, HCI Across Borders (HCIxB) workshop at CHI	*New Orleans, USA*
2021	**Lead Organizer**, HCI Across Borders (HCIxB) workshop at CHI	*Yokohama, Japan*
2020	**Organizer**, HCI Across Borders (HCIxB) symposium at CHI	*Hawaii, USA*
2019	**Organizer**, HCI Across Borders (HCIxB) symposium at CHI	*Glasgow, UK*
2019	**Organizer**, Young Researchers Reception at ICTD	*Ahmedabad, India*
2020–2022	**Executive Committee Member**, SIGCHI Development Fund	

Program Committees and Reviewing

2024–	**Program Committee**, ACM COMPASS 2025, AfriCHI 2025, ICTD 2024
2020–2023	**Associate Chair**, CHI Extended Abstracts
2018–	**Reviewer (80+ papers)**, CHI, CSCW, ACM COMPASS, ICTD, CHI Case Studies

Professional Experience

Carnegie Mellon University, Africa — *Kigali, Rwanda*
Visiting Researcher — *2018–2020 (summers, spring)*
- Host: Prof. Tim Brown
- Project: Design of a Phone Call-Based Health Information System
- Conducted participatory research with rural women and prototyped an IVR system in rural Rwanda

United Nations University Computing and Society — *Macau, SAR, China*
Research Intern — *Jul 2017 – Sep 2017*
- Advisor: Dr. Mamello Thinyane
- Conducted participatory design with a local NGO
- Built a mobile app to report and monitor homelessness in Macau

Carnegie Mellon University — *Pittsburgh, USA*
Senior Web Developer — *Jul 2015 – Jun 2017*
- Advisor: Prof. Geoff Gordon
- Built and managed educational technology to support online learning
- Prototyped a low-cost video recording system for teaching courses

Carnegie Mellon University — *Pittsburgh, USA*
Research Assistant — *Aug 2015 – Jun 2017*
- Advisor: Prof. Jen Mankoff
- Longitudinal eating detection: deployed and evaluated the system with 12 users in the field
- Twist & Knock: a one-handed gesture for smartwatches (S1)
- EDigs: built and maintained the APIs for an apartment-hunting website in Pittsburgh

TeliportMe Inc: Indian startup — *Karnataka, IN*

Web Architect — *Jul 2011 – Aug 2014*

- Engineered APIs for Android, iOS, web, and Google Glass
- Scaled back-end to handle 4 million users and manage 100K locations
- Designed a 3D viewer on par with Google Street View
- Managed 10 Ubuntu machines with MySQL and Redis replication

Skills

Methods	Participatory design, longitudinal field deployment, interviews, think-aloud, mixed-methods analysis
Programming	Python, JavaScript, PHP, HTML/CSS, Android, iOS, Arduino
Tools and Platforms	LLM APIs, chatbot and IVR platforms, WhatsApp deployment, pandas, scikit-learn, MySQL, Redis

Selected Press Coverage

The Next Web — *TeliportMe introduces 3D effects to its 360-degree panoramic app for Android*
TechCrunch — *Test Flighting: Applover Launches Free, Crowdsourced Testing Platform for Android Apps*
Robert Scoble — *Developers: crowd-source your testing of Android phones*
CMU — *A new online resource from Carnegie Mellon University seeks to help people find the perfect rental*
CMU HCII — *EDigs Helps Apartment Hunters Find the Perfect Rental*
The Better India — *How a National Level Cyclist Is Fighting Both Spinal Injury & Poverty with Inspiring Confidence*
Milaap Blog — *Getting Back to Cycling: A National Medalist's Dream*

Beyond Research

- **Philanthropy** — Raised over **$20,000** for causes in India through swimming and crowdfunding
- **Android** — Shipped 3 apps to the Play Store, with roughly 10,000 active users
- **Web** — Deployed 20+ websites to production; the largest reached over **4 million users**
- **Teaching** — Taught Kannada as a second language to children at the S.V. Temple in Pittsburgh
- **Swimming** — Swam 10 miles from Elephanta Caves to the Gateway of India to raise money for acid-attack survivors